

Neural Familiarity Classifier

Neuroinformatics

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Overview & Methods

- **Goal:** Classify whether a stimulus is familiar or new from multi-neuron spike-trains.
- **Data:** 800 trials $\times$ 1000 neurons $\times$ 501 time bins; binary labels (familiar = 1, new = 0).
- **Feature extraction:**
 - Average firing rate per neuron
 - Neural synchrony (mean pairwise correlation)
 - Population spectral power (mean FFT amplitude)
- **Models:** Logistic Regression, Random Forest, SVM
- **Evaluation:** Stratified 5-fold CV, ROC AUC primary metric

Raw Data Visualization

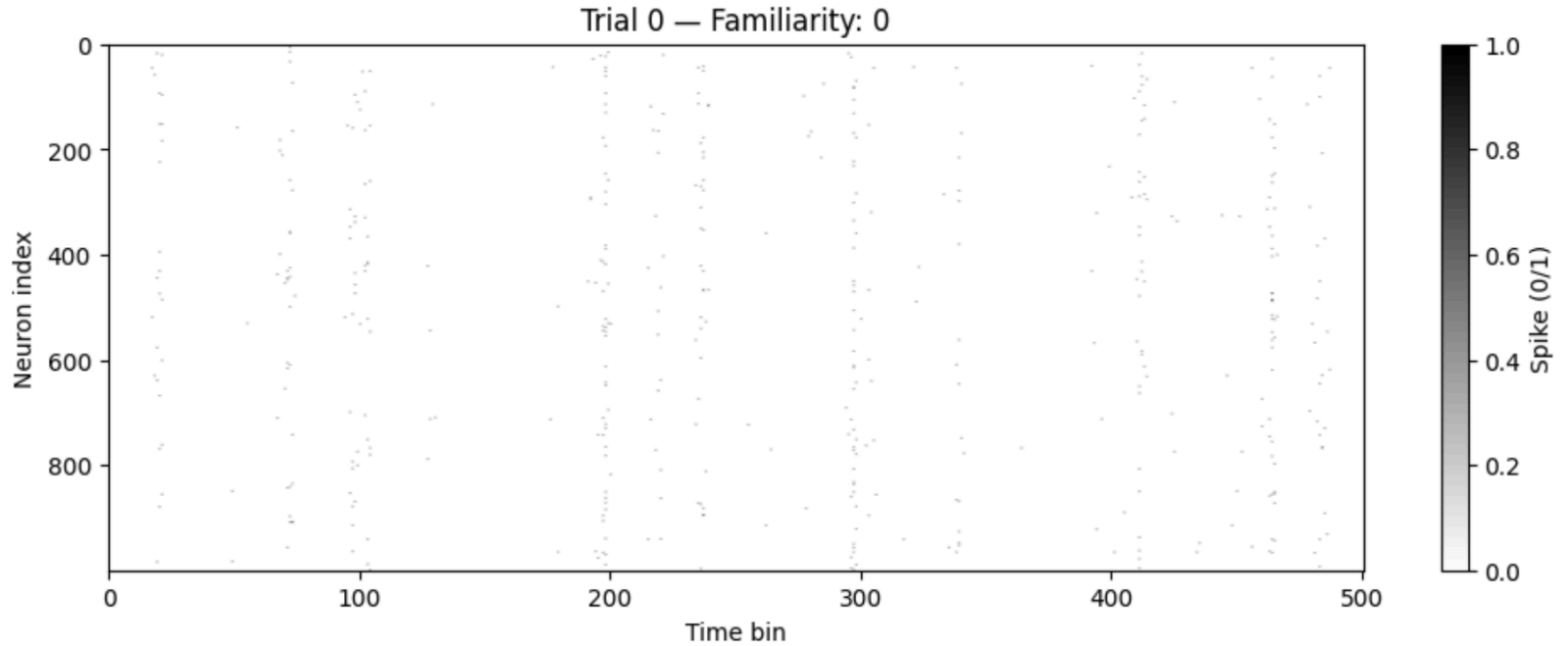
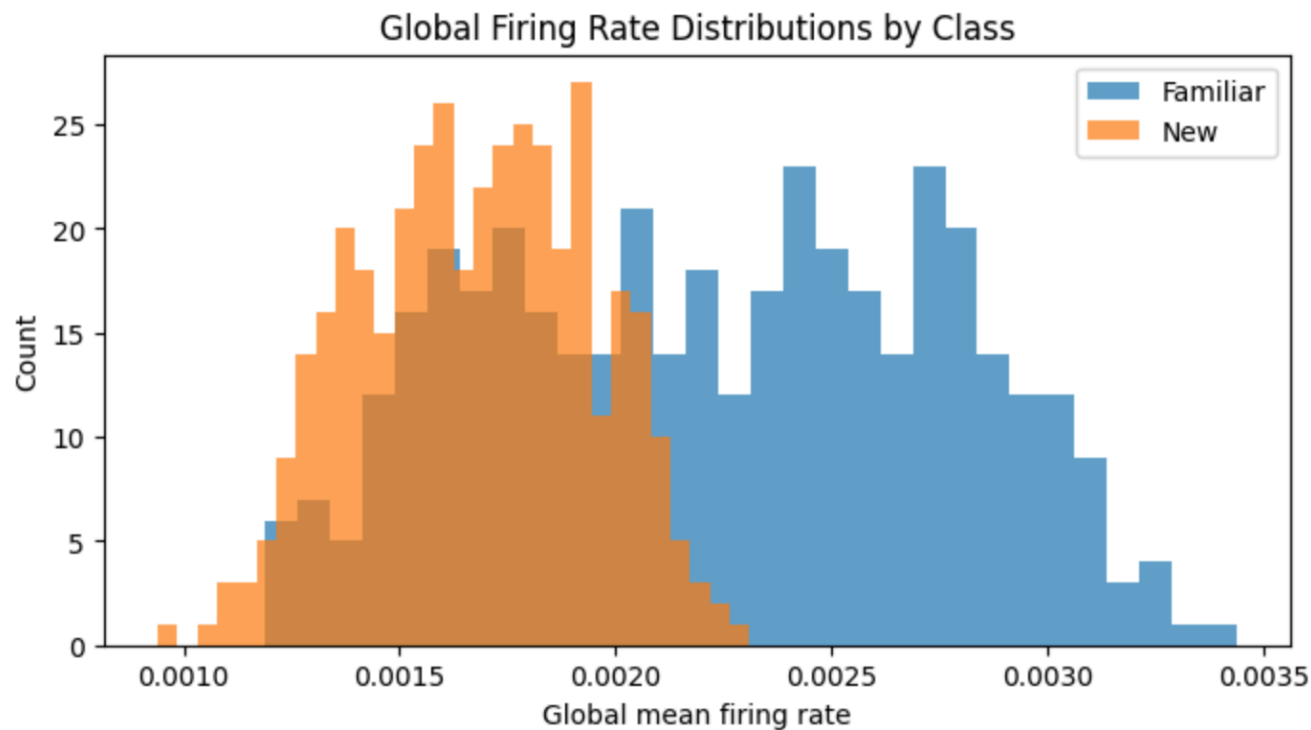


Fig01: Raster plot of Trial 0 (dark = spike, light = silence), showing how neurons respond over time to a familiar stimulus.



KS statistic = 0.542, p-value = 2.456e-54
KL divergence (familiar || new) = 11.459
KL divergence (new || familiar) = 0.808

Fig02: Significant shift in firing-rate distributions suggests firing rate is informative of familiarity.

	Features	Model	Mean AUC	Std AUC
0	Rate	LogReg	0.902	0.037
1	Rate	RandomForest	0.976	0.035
2	Rate	SVM	0.992	0.014
3	Combined	LogReg	0.788	0.177
4	Combined	RandomForest	0.979	0.038
5	Combined	SVM	0.777	0.181

Fig03: Mean ROC AUC (with ± 1 SD error bars) across feature sets and classifiers.

-> Random Forest on Combined features achieves the highest AUC, demonstrating the added value of synchrony and spectral-power features and the power of a non-linear model.

Conclusions

- **Firing-rate coding works:** Global mean firing rate alone gives $AUC \approx 0.81$.
- **Richer features help:** Adding synchrony & spectral power boosts AUC to ≈ 0.98 with Random Forest.
- **Non-linear modeling wins:** Random Forest outperforms Logistic Regression and SVM, capturing complex interactions.

Limitations

- Synchrony feature required NaN imputation for silent neurons.
- We only tested on single session data—generalization to new sessions/animals unknown.
- FFT power collapses time resolution; finer temporal codes (e.g. wavelets) could add information.

Next Steps

1. **Hyperparameter tuning** (e.g. RF depth, SVM kernels) via GridSearchCV for optimal settings.
2. **Temporal features:** Extract inter-spike-interval statistics or time-resolved rate to capture dynamic patterns.
3. **Cross-session validation:** Train/test on different recording days or subjects to assess robustness.
4. **Feature importance analysis:** Use SHAP or permutation importance to identify which neurons or features drive familiarity.